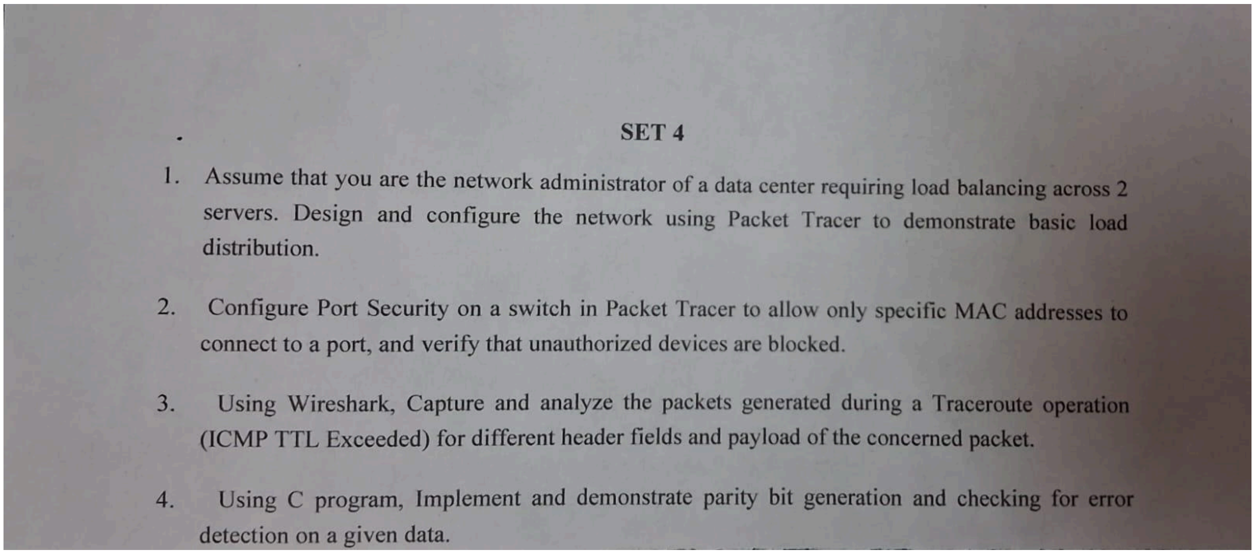
MODEL PRACTICAL EXAM QUESTION PAPER:



SET 4

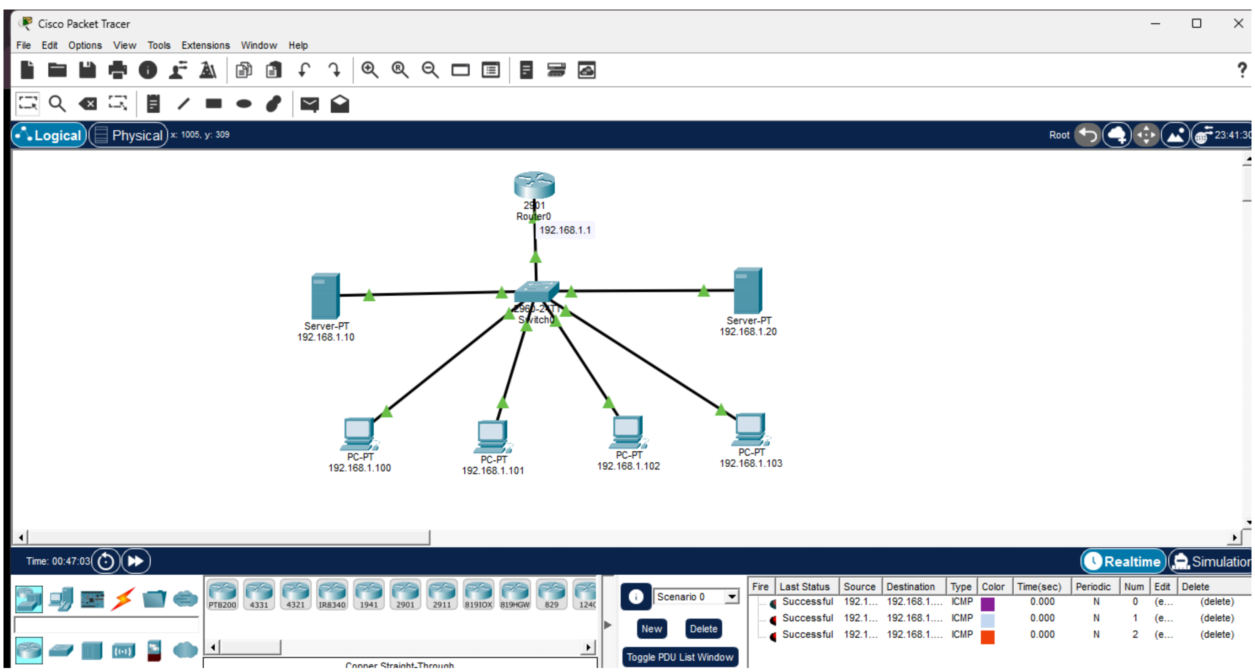
1. Assume that you are the network administrator of a data center requiring load balancing across 2 servers. Design and configure the network using Packet Tracer to demonstrate basic load distribution.

2. Configure Port Security on a switch in Packet Tracer to allow only specific MAC addresses to connect to a port, and verify that unauthorized devices are blocked.

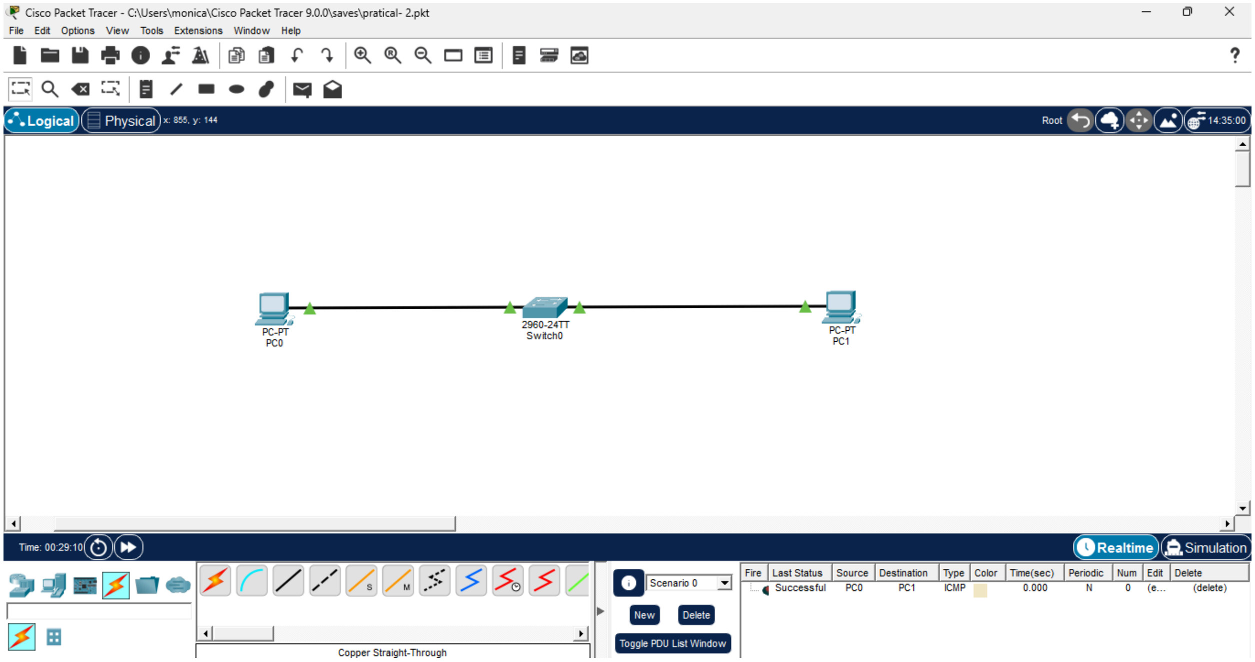
3. Using Wireshark, Capture and analyze the packets generated during a Traceroute operation (ICMP TTL Exceeded) for different header fields and payload of the concerned packet.

4. Using C program, Implement and demonstrate parity bit generation and checking for error detection on a given data.

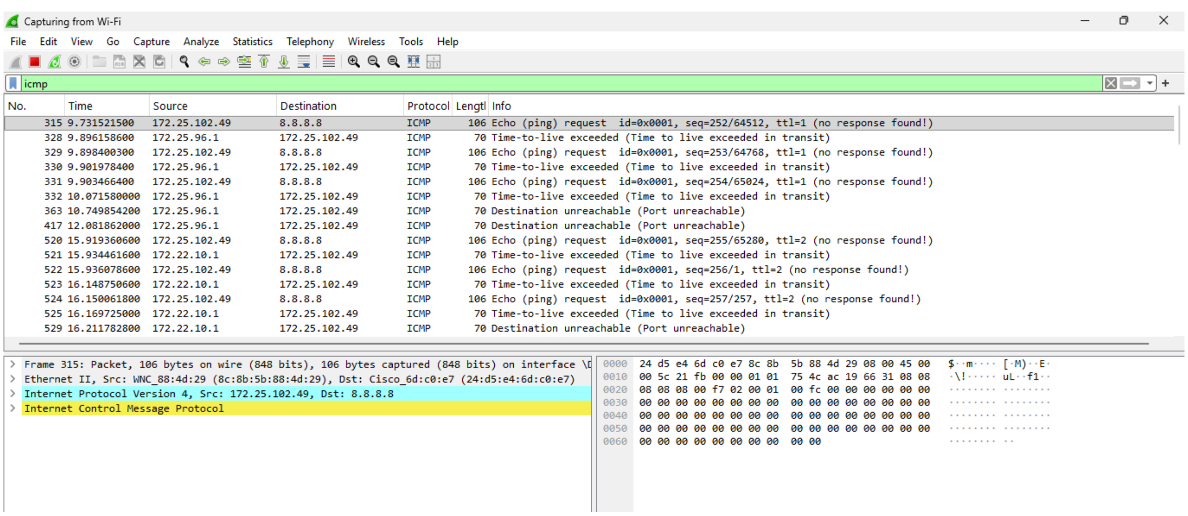
MODEL PRACTICAL EXAM
QUESTION 1:



QUESTION - 2



QUESTION 3:



```
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\monica>tracert 8.8.8.8

Tracing route to dns.google [8.8.8.8]
over a maximum of 30 hops:

  1   164 ms     3 ms   168 ms  172.25.96.1
  2    15 ms   212 ms    19 ms  172.22.10.1
  3    24 ms    49 ms     6 ms  14.139.187.225
  4     *        *        *     Request timed out.
  5     *        *        *     Request timed out.
  6     *        *        *     Request timed out.
  7     *        *        *     Request timed out.
  8    86 ms     9 ms    20 ms  72.14.213.20
  9    48 ms   100 ms    26 ms  216.239.43.133
 10   133 ms    37 ms   113 ms  142.251.55.43
 11   246 ms    26 ms    25 ms  dns.google [8.8.8.8]

Trace complete.

C:\Users\monica>
```

QUESTION 4:

```c
    if (parity == 0)
        strcat(transmitted, "0");
    else
        strcat(transmitted, "1");

    printf("Transmitted data = %s\n", transmitted);

    printf("Enter position to introduce error (1-%d): ",
           (int)strlen(transmitted));
    scanf("%d", &error_position);

    if (error_position >= 1 &&
        error_position <= strlen(transmitted)) {

        if (transmitted[error_position - 1] == '0')
            transmitted[error_position - 1] = '1';
        else
            transmitted[error_position - 1] = '0';
    }

    printf("Received data = %s\n", transmitted);

    count = 0;
```

```
Enter position to introduce error (1-8): 3
Received data = 10010010
Error detected in received data.


...Program finished with exit code 0
Press ENTER to exit console.
```

Model Practical Exam:

1. Load Balancing Across Two Servers

Aim:

To design and configure a network with two servers and demonstrate basic load balancing using Cisco Packet Tracer.

Required Materials:

Cisco Packet Tracer, 1 router, 1 switch, 2 servers, PCs, and network cables.

Procedure:

1. Connect the router, switch, two servers, and PCs
2. Assign suitable IP Address to all devices.
3. Configure the router as the default gateway.
4. Configure both servers to handle client requests.
5. Distribute requests between the two servers and test the network.

Result:

The network was successfully configured, and client requests was distributed between the two servers, demonstrating load balancing.

2. Port Security on a Switch :-

Aim:

To configure port security on a switch and prevent unauthorized devices from accessing the network.

Required Network:

Cisco Packet Tracer, 1 switch, 2 PCs, and Networking cables.

Procedure:

1. Connect the PCs to the switch.
2. Configure the required switch port as an access port
3. Enable port security and specify the ~~am~~ authorized MAC addres
4. Set the violation mode to shutdown.
5. Connect an unauthorized device and observe the port status.

Result:

Port Security was successfully configured. The authorized device was allowed access, while the unauthorized device was blocked.

5) ICMP TTL Exceeded Using Wireshark.

Aim:

To capture and analyze ICMP Time-to-Live (TTL) Exceeded packets using Wireshark.

Required Materials :

Computer /PC , Wireshark , Internet connection , and command prompt.

Produre :

1. Open Wireshark and start capturing packets from the Wi-Fi interface
2. Open Command Prompt and execute tracert 8.8.8.8.
3. Stop the packet capture after the Traceroute is completed.
4. Apply the icmp filter in Wireshark.
5. Identify and analyze the Time-tolive exceeded packets.

Result:

ICMP TTL Exceeded packets were successfully captured and analyzed, showing the intermediate routers in the network path.

4) Parity Bit Generation and Error Detection Using C.

Aim:

To develop a C program for generating an even parity bit and detecting errors in transmitted binary data.

Required Materials:

Computer, VS code (or) C compiler, and C programming environment.

Procedure:

1. Enter the binary data as input.
2. Count the number of 1s and generate the even parity bit.
3. Append the parity bit to the data.
4. Introduce an error by changing one bit.
5. Check the received data and determine whether an error has occured.

Result:

The C program successfully generated the parity bit and detected the error in the received binary data.